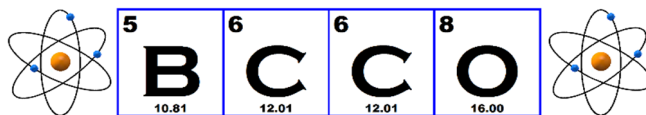


2025 JUNIOR CANADIAN CHEMISTRY OLYMPIAD



The Periodic Table/Data Sheet may be used for all parts of the contest, and non-programmable calculators are allowed.

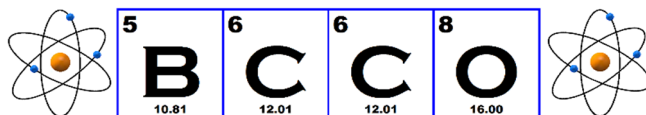
考试期间允许使用化学元素周期表、数据表以及非编程计算器。

Write your answers on the Answer Sheet provided using a 2B pencil.

请将答案用2B 铅笔正确填涂在答题卡上。

Students are expected to attempt all questions for a complete examination in 1.5 hours.

考试时长为 1.5 小时。



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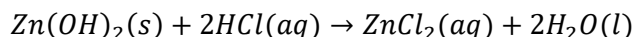
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考试时长为 1.5 小时。

- 1) What is the type of the following reaction?

以下反应属于哪种类型？



- | | |
|-----------------------|------|
| A) Synthesis | 合成反应 |
| B) Single replacement | 置换反应 |
| C) Decomposition | 分解反应 |
| D) Neutralization | 中和反应 |
| E) Combustion | 燃烧反应 |

- 2) The density of zinc chloride at 20 °C is 2.907 g/cm³. What is the volume of 1 mole of zinc chloride at 20 °C?

温度为 20°C 时，氯化锌的密度为 2.907 g/cm³。请问温度为 20°C 时，1 摩尔氯化锌的体积是多少？

- A) 59.1 cm³ B) 34.7 cm³ C) 46.9 cm³ D) 136 cm³ E) 396 cm³

- 3) Which of the following equations is not properly balanced?

以下哪个方程式没有正确配平？

- A) $2\text{CH}_3\text{OH} + 3\text{O}_2 \rightarrow 2\text{CO}_2 + 4\text{H}_2\text{O}$
 B) $3\text{NaOH} + \text{H}_3\text{PO}_4 \rightarrow \text{Na}_3\text{PO}_4 + 2\text{H}_2\text{O}$
 C) $\text{C}_{12}\text{H}_{22}\text{O}_{11} + \text{H}_2\text{O} \rightarrow 2\text{C}_6\text{H}_{12}\text{O}_6$
 D) $4\text{C}_3\text{H}_5\text{N}_3\text{O}_9 \rightarrow 12\text{CO}_2 + 10\text{H}_2\text{O} + 6\text{N}_2 + \text{O}_2$
 E) $4\text{NH}_3 + 3\text{O}_2 \rightarrow 6\text{H}_2\text{O} + 2\text{N}_2$

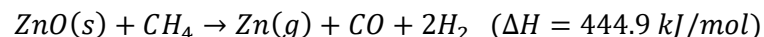
- 4) Potassium permanganate (KMnO₄) is dark purple, almost black. It is an oxidising agent with disinfectant, deodorising, and astringent properties. How many grams of potassium permanganate is required to prepare a 0.1 M solution in a 200 mL volumetric flask?

高锰酸钾 (KMnO₄) 呈近乎黑色的深紫色，它是一种具有消毒、除臭和止血的功能的氧化剂。要在一个 200 mL 容量瓶中配置 0.1 M 的溶液，需要多少克高锰酸钾？

- A) 0.316 B) 15.8 C) 2.20 D) 11.0 E) 3.16

- 5) Solar energy is a plentiful, clean, and sustainable energy source, which can contribute to protect the environment. Hydrogen is also an important sustainable energy option, but clean hydrogen production can be problematic. 50% of the world's hydrogen production is done by methane reforming. Methane reforming is a chemical reaction consisting in extracting dihydrogen from methane, the reaction is endothermic and therefore requires heat. A promising method of methane reforming uses a solar-driven approach in combination with zinc oxide:

太阳能是一种丰富、清洁且可持续的能源，有助于保护环境。氢能也是一种重要的可持续能源，但清洁氢的制造过程可能是个难题。全球 50% 的氢通过甲烷重整制备，这是一种从甲烷中提取氢气的吸热化学反应，所以需要外部提供热能。一种前景较好的甲烷重整方法，在结合氧化锌时采用了太阳能驱动：



If a total of 87 million tons of dihydrogen are produced in one year, how many million of tons of carbon monoxide are produced? Assume that methane reforming is the sole source of carbon monoxide.

若全年共生产 8700 万吨氢气，则同期会产生几百万吨一氧化碳？(假设一氧化碳的唯一来源是甲烷重整)

- A) 302 B) 43.5 C) 174 D) 87 E) 604

- 6) When a colorless solution A is added to a sodium chloride solution, a white precipitate is formed. The precipitate is soluble in ammonia. The electrolysis of A gives a shiny metal that is a very good electricity conductor. Identify the metal in solution A.

向氯化钠溶液中加入无色溶液 A 时，会形成白色沉淀，该沉淀可溶于氨水。对溶液 A 进行电解，能得到一种带有金属光泽、导电性良好的金属，请问溶液 A 中的金属元素是什么？

- A) Na B) Fe C) Al D) Ag E) Cu

- 7) In UV/Vis spectroscopy, the Beer-Lambert law equation relates the absorbance, the molar absorptivity of the molecule, the path length of the light, and the concentration of the solution as follows:

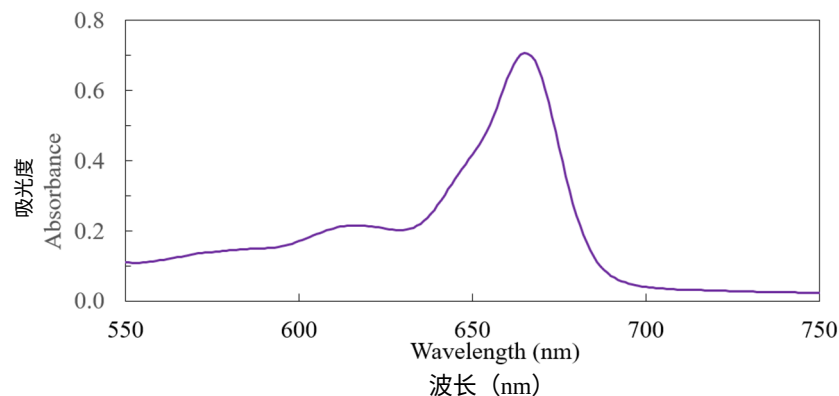
在紫外可见分光光度法 (UV/Vis) 中，朗伯-比尔定律 (Beer-Lambert law) 方程将吸光度、分子的摩尔吸光系数、光的路径长度以及溶液浓度之间的关系表述如下：

$$A = \epsilon lc$$

Where A is absorbance, ϵ is the molar extinction coefficient (a property of the molecule), and l is the path length of the light through the sample, and c the concentration of the solution. A cell with an optical path of 1 cm was filled with a solution of chlorophyll a in acetone.

The molar extinction coefficient of chlorophyll a is $7.9 \times 10^4 \text{ M}^{-1} \text{ cm}^{-1}$ at 665 nm. The spectrum recorded is shown below. What is the concentration of chlorophyll a ?

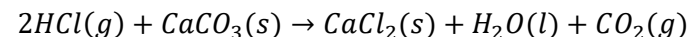
其中 A 为吸光度, ϵ 为摩尔消光系数 (分子的一种固有属性), l 为光线通过样品的路径长度, c 为溶液浓度。现使用光的路径长度为 1 cm 的比色皿, 注入溶于丙酮的叶绿素 a 溶液。叶绿素 a 在 665 nm 波长下的摩尔消光系数为 $7.9 \times 10^4 \text{ M}^{-1} \text{ cm}^{-1}$ 。测得光谱如下图所示, 请问叶绿素 a 的浓度是多少?



- A) $1.33 \times 10^{-8} \text{ mol/L}$
 B) $8.86 \times 10^{-6} \text{ mol/L}$
 C) $1.27 \times 10^{-5} \text{ mol/L}$
 D) $8.42 \times 10^{-3} \text{ mol/L}$
 E) $1.05 \times 10^{-3} \text{ mol/L}$
- 8) The Athabasca basin, located in northern Saskatchewan and Alberta, provides the highest grade of uranium ore in the world. The region supplies about 20% of the world's uranium. The half-life of uranium-238 is about 4.5 billion years. What is the rate constant for the uranium-238 decay?
- 位于萨斯喀彻温省和阿尔伯塔省北部的阿萨巴斯卡盆地, 拥有全球最高品位的铀矿石。全球约 20% 的铀产自该地区。铀-238 的半衰期约为 45 亿年, 其衰变速率常数是多少?
- A) $2.9 \times 10^{-16} \text{ s}^{-1}$ B) $4.2 \times 10^{-13} \text{ day}^{-1}$ C) 0.15 yr^{-1}
 D) $4.2 \times 10^{-4} \text{ day}^{-1}$ E) $2.2 \times 10^{-10} \text{ yr}^{-1}$

- 9) In a 1 L glass container, solid calcium carbonate reacts with hydrogen chloride gas as follows:

在一个 1 L 的玻璃容器中, 固体碳酸钙与氯化氢气体发生如下反应:



Initially the glass container is filled with 10 atm of HCl gas and 10 g of calcium carbonate only. Select the statement that is true if this reaction is carried at 25°C.

初始时, 玻璃容器内仅充有 10 个标准大气压 (atm) 的氯化氢气体和 10 g 碳酸钙。若反应在 25°C 下进行, 请判断下列哪项陈述正确。

- A) HCl is the limiting reagent.
氯化氢 (HCl) 是限制反应物
- B) 22.2 g of CaCl_2 will be produced.
将生成 22.2 g 氯化钙 (CaCl_2)
- C) The final pressure will be higher than the initial pressure.
最终压强将高于初始压强
- D) CaCO_3 is the limiting reagent.
碳酸钙 (CaCO_3) 是限制反应物
- E) 0.198 moles of CO_2 will be produced.
将生成 0.198 摩尔二氧化碳 (CO_2)
- 10) In assuming ideal gas behavior for the reaction in question 9, what would be the final pressure in the container at 25°C?
- 假设题目 9 中参加反应的气体为理想气体, 温度为 25°C 时容器内的最终压强会是多少?
- A) 7.58 atm B) 2.42 atm C) 0 atm D) 5.15 atm E) 12.4 atm

- 11) Effective nuclear charge (Z_{eff}) is a useful tool for understanding periodicity. Z_{eff} is the net positive charge experienced by an electron in a polyelectronic atom. The term "effective" is used because the positive charge of the nuclei cannot reach a given electron completely due to the shielding effect of other electrons. Z_{eff} is a theoretically calculated value. A good estimate of Z_{eff} can be obtained using Slater's rule. Slater's rule states that $Z_{\text{eff}} = Z - S$, where Z is the atomic number of the atom and S a screening constant. The screening value is 0.85 for a core electron and 0.35 for a valence electron. The screening constant (S) is obtained by the summation of all the screening values of other electrons, excluding the electron for which the effective nuclear charge is to be determined. Use Slater's rule to estimate the Z_{eff} for a valence electron in a hydride anion (H^-).

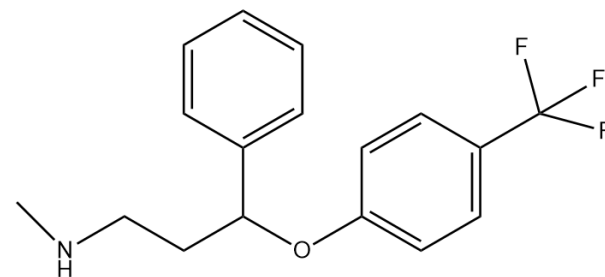
有效核电荷 (Z_{eff}) 是理解元素周期律的一个有力工具。 Z_{eff} 指的是多电子原子中，某一个电子所感受到的净正电荷。之所以使用“有效”一词，是因为受其他电子的屏蔽效应影响，原子核的正电荷无法完全作用于该特定的电子。 Z_{eff} 是一个理论计算值，可以利用斯莱特规则 (Slater's rule) 进行估算。斯莱特规则指出： $Z_{\text{eff}} = Z - S$ ，其中 Z 为原子的原子序数， S 为屏蔽常数。内层电子的屏蔽值为 0.85；价电子的屏蔽值为 0.35。屏蔽常数 (S) 等于除待求有效核电荷的电子之外，其他所有电子的屏蔽值之和。请使用斯莱特规则估算氢负离子 (H^-) 中一个价电子的 Z_{eff} 值。

- A) 1.00 B) 0.15 C) 1.65 D) -1.00 E) 0.65

- 12) The structure of an antidepressant that is on the World Health Organisation's List of Essential Medicines is shown below. This Medicine is prescribed to more than 24 million people in the USA. Which of the following molecular formula is the correct one for the antidepressant shown below?

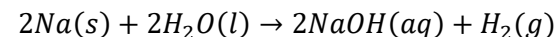
下图展示了一种抗抑郁药的结构，该药物已被列入世界卫生组织 (WHO) 的《基本药物清单》。在美国，医生已为超过 2400 万名患者开具该药物。下列哪项是下图所示抗抑郁药的正确分子式？

- A) $\text{C}_{17}\text{H}_{17}\text{F}_3\text{NO}$
B) $\text{C}_{17}\text{H}_{18}\text{F}_3\text{NO}$
C) $\text{C}_{16}\text{H}_{15}\text{F}_3\text{NO}$
D) $\text{C}_{18}\text{H}_{18}\text{F}_3\text{NO}$
E) $\text{C}_{18}\text{H}_{17}\text{F}_3\text{NO}$



- 13) A piece of sodium is added to water in a beaker and fire can be observed. The reaction that takes place is:

将一块钠放入盛有水的烧杯中，可观察到火焰。
此时发生的反应是：



Predict the signs of ΔH , w and ΔU . (Answers given in this order).

判断 ΔH 、 w 和 ΔU 的符号 (答案按此顺序给出)。

- A) - / - / -
B) + / - / +
C) - / + / +
D) + / + / +
E) - / - / +

- 14) A diamond is the hardest natural substance on earth, but if it is placed in an oven and the temperature is raised to about 763°C , it will simply vanish, without even ash remaining. Only a little carbon dioxide will have been released. If the heats of combustion of diamond and graphite are -395.4 kJ/mol and -393.5 kJ/mol respectively, what is the heat of formation of diamonds?

金刚石是地球上最坚硬的天然物质，但如果将其置于炉中并将温度升至约 763°C ，它将完全消失，甚至不留灰烬。该过程仅释放出少量的二氧化碳。如果金刚石和石墨的燃烧热分别为 -395.4 kJ/mol 和 -393.5 kJ/mol ，那么金刚石的生成焓是多少？

- A) -1900 J/mol B) 788.9 kJ/mol C) -788.9 kJ/mol
D) -0.002 kJ/mol E) 1900 J/mol

- 15) How much energy is needed to heat 30mL of water from 5° to 30°C ?
($C_{p,s} = 4.18\text{ Jg}^{-1}\text{K}^{-1}$)
将 30mL 水从 5°C 加热到 30°C 需要多少能量？($C_{p,s} = 4.18\text{ Jg}^{-1}\text{K}^{-1}$)
- A) 1881 J B) 174.2 J C) 3135 J D) 104.5 J E) 38000 J

- 16) Following the VSEPR theory, which of the following molecules has a molecular geometry that is trigonal bipyramidal?

根据价层电子对互斥理论 (VSEPR theory)，下列哪种分子的分子几何构型为三角双锥形？

- A) ClF_3 B) PF_2Cl_3 C) NH_3 D) XeOF_4 E) BrF_5

- 17) Which of the following atoms has the highest first ionization energy?
下列原子中哪一个的第一电离能最高？

- A) F B) He C) Cl D) Ar E) Br

- 18) Which of the following species has the smallest radius?
下列哪个粒子的半径最小？

- A) H^- B) He C) Li^+ D) Be^{2+} E) B^{3+}

- 19) Which of the following double replacement reactions will form a yellow precipitate?

下列哪一个复分解反应会生成黄色沉淀？

- A) $\text{AgNO}_3(\text{aq}) + \text{NaCl}(\text{aq})$
B) $\text{AgNO}_3(\text{aq}) + \text{KI}(\text{aq})$
C) $\text{AgNO}_3(\text{aq}) + \text{NaCH}_3\text{COO}(\text{aq})$
D) $\text{AgNO}_3(\text{aq}) + \text{K}_2\text{CrO}_4(\text{aq})$
E) $\text{AgNO}_3(\text{aq}) + \text{K}_2\text{SO}_4(\text{aq})$

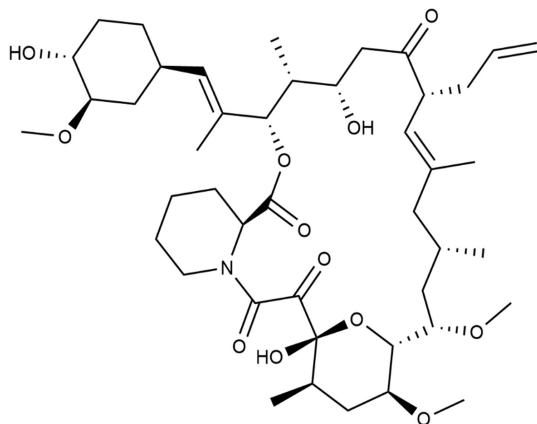
- 20) Dimethyl ether and ethanol are organic isomers, they have the same
二甲醚和乙醇属于有机同分异构体，二者具有相同的

- | | |
|--|------------------------------------|
| I. Molar mass. | 摩尔质量 |
| II. Normal boiling point. | 标准沸点 |
| III. Surface tension. | 表面张力 |
| IV. Density at 20°C and 1 atm. | 20°C 、1个标准大气压下的密度 |
| V. Density at 100°C and 1 atm. | 100°C 、1个标准大气压下的密度 |

- A) I and / 和 II B) I and / 和 III C) I and / 和 IV
D) I and / 和 V E) I, IV and / 和 V

- 21) Tacrolimus is a drug used after organ transplant to lower the risk of organ rejection. The structure is shown below. Which of the following functional groups the molecule contains?

他克莫司 (Tacrolimus) 是一种用于器官移植后降低排异风险的药物, 其结构如下图所示。该分子含有以下哪些官能团?



- A) Alkene, alcohol, amine, and ketone 烯烃、醇、胺和酮
 B) Alkyne, alcohol, amine, and ether 炔烃、醇、胺和醚
 C) Alkene, amide, ester, and ketone 烯烃、酰胺、酯和酮
 D) Aldehyde, amide, ester, and ketone 醛、酰胺、酯和酮
 E) Alkene, amine, ether, and ester 烯烃、胺、醚和酯
- 22) Each of the following 1 liter of aqueous solution is prepared by dissolving 0.1 moles of each of the compounds in a sufficient amount of water. Which solution will have the highest percentage of free ions for all ions both at pH = 0 and pH = 14?
- 将 0.1 摩尔的下列化合物分别溶于足量水中, 各配制成 1 升水溶液。请问哪一种溶液在 pH = 0 和 pH = 14 两种条件下, 其所有离子的游离百分比均为最高?

- A) NaCl and KNO₃ NaCl 和 KNO₃
 B) NH₄Cl and NaCH₃COO NH₄Cl 和 NaCH₃COO
 C) CuSO₄ and NaCl CuSO₄ 和 NaCl
 D) Na₂S and NaCl Na₂S 和 NaCl
 E) Na₂CO₃ and Ca(NO₃)₂ Na₂CO₃ 和 Ca(NO₃)₂

- 23) Which of the following is a valid set of quantum numbers for the unpaired electron in ground state copper atom?

下列哪一组是基态铜原子中未成对电子的有效量子数组合?

- A) 3, 1, 0, $-\frac{1}{2}$ B) 3, 2, 2, $\frac{1}{2}$ C) 3, 3, -1, $-\frac{1}{2}$
 D) 4, 0, 0, $\frac{1}{2}$ E) 4, 1, 0, $\frac{1}{2}$

- 24) Thermal decomposition of a sample of 1.25 grams of HgO containing unreactive impurities produces 56.35mL of oxygen gas collected at 22°C under standard atmospheric pressure. What is the percent purity of the sample? (Hint, the products are mercury and oxygen gas).

对一份 1.25 克且含有惰性杂质的氧化汞样品进行热分解, 在 22°C 和标准大气压下收集到 56.35mL 氧气。求该样品的纯度百分比是多少? (提示: 反应产物为汞和氧气)

- A) 40.3% B) 49.2% C) 89.2%
 D) 80.6% E) 98.4%

- 25) Metalloid elements are key to the microelectronics industry. Microchips used in microelectronics are made by silicon, germanium and other metalloid elements. N-type semiconductors are made by

准金属元素对微电子工业至关重要。该工业中使用的微芯片是由硅、锗和其他准金属元素制成的。N 型半导体是通过下列哪种方式制备的?

- A) Doping Silicon with Germanium 用锗掺杂硅
 B) Doping Silicon with Arsenic 用砷掺杂硅
 C) Doping Silicon with Boron 用硼掺杂硅
 D) Doping Arsenic with Silicon 用硅掺杂砷
 E) Doping Germanium with Silicon 用硅掺杂锗

End of examination questions.

1 H 1.008 Hydrogen																	2 He 4.003 Helium
3 Li 6.941 Lithium	4 Be 9.012 Beryllium	JUNIOR CANADIAN CHEMISTRY OLYMPIAD										5 B 10.81 Boron	6 C 12.01 Carbon	7 N 14.01 Nitrogen	8 O 16.00 Oxygen	9 F 19.00 Fluorine	10 Ne 20.18 Neon
11 Na 22.99 Sodium	12 Mg 24.31 Magnesium	Periodic table and data sheet										13 Al 26.98 Aluminium	14 Si 28.09 Silicon	15 P 30.97 Phosphorus	16 S 32.07 Sulfur	17 Cl 35.45 Chlorine	18 Ar 39.95 Argon
19 K 39.10 Potassium	20 Ca 40.08 Calcium	21 Sc 44.96 Scandium	22 Ti 47.87 Titanium	23 V 50.94 Vanadium	24 Cr 52.00 Chromium	25 Mn 54.94 Manganese	26 Fe 55.85 Iron	27 Co 58.93 Cobalt	28 Ni 58.69 Nickel	29 Cu 63.55 Copper	30 Zn 65.38 Zinc	31 Ga 69.72 Gallium	32 Ge 72.61 Germanium	33 As 74.92 Arsenic	34 Se 78.96 Selenium	35 Br 79.90 Bromine	36 Kr 83.80 Krypton
37 Rb 85.47 Rubidium	38 Sr 87.62 Strontium	39 Y 88.89 yttrium	40 Zr 91.22 Zirconium	41 Nb 92.91 Niobium	42 Mo 95.96 Molybdenum	43 Tc (98) Technetium	44 Ru 101.1 Ruthenium	45 Rh 102.9 Rhodium	46 Pd 106.4 Palladium	47 Ag 107.9 Silver	48 Cd 112.4 Cadmium	49 In 114.8 Indium	50 Sn 118.7 Tin	51 Sb 121.8 Antimony	52 Te 127.6 Tellurium	53 I 126.9 Iodine	54 Xe 131.3 Xenon
55 Cs 132.9 Cesium	56 Ba 137.3 Barium	57 La 138.9 Lanthanum	72 Hf 178.5 Hafnium	73 Ta 180.9 Tantalum	74 W 183.9 Tungsten	75 Re 186.2 Rhenium	76 Os 190.2 Osmium	77 Ir 192.2 Iridium	78 Pt 195.1 Platinum	79 Au 197.0 Gold	80 Hg 200.6 Mercury	81 Tl 204.4 Thallium	82 Pb 207.2 Lead	83 Bi 209.0 Bismuth	84 Po (209) Polonium	85 At (210) Astatine	86 Rn (222) Radon
87 Fr (223) Francium	88 Ra (226) Radium	89 Ac (227) Actinium	104 Rf (261) Rutherfordium	105 Db (262) Dubnium	106 Sg (266) Seaborgium	107 Bh (264) Bohrium	108 Hs (277) Hassium	109 Mt (268) Meitnerium	110 Ds (269) Darmstadtium	111 Rg (272) Roentgenium	112 Cn (285) Copernicium	113 Nh (284) Nihonium	114 Fl (289) Flerovium	115 Mc (288) Moscovium	116 Lv (292) Livermorium	117 Ts (294) Tennessine	118 Og (294) Oganesson

58 Ce 140.1 Cerium	59 Pr 140.9 Praseodymium	60 Nd 144.2 Neodymium	61 Pm (145) Promethium	62 Sm 150.4 Samarium	63 Eu 152.0 Europium	64 Gd 157.3 Gadolinium	65 Tb 158.9 Terbium	66 Dy 162.5 Dysprosium	67 Ho 164.9 Holmium	68 Er 167.3 Erbium	69 Tm 168.9 Thulium	70 Yb 173.0 Ytterbium	71 Lu 175.0 Lutetium
90 Th 232.0 Thorium	91 Pa 231.0 Protactinium	92 U 238.0 Uranium	93 Np 237.0 Neptunium	94 Pu (244) Plutonium	95 Am (243) Americium	96 Cm (247) Curium	97 Bk (247) Berkelium	98 Cf (251) Californium	99 Es (252) Einsteinium	100 Fm (257) Fermium	101 Md (258) Mendelevium	102 No (259) Nobelium	103 Lr (262) Lawrencium

原子质量单位	Atomic mass unit	1.66054×10 ⁻²⁷ kg
阿伏伽德罗常数	Avogadro's number	6.022×10 ²³
电子电荷量	Charge of an electron	1.60218×10 ⁻¹⁹ C
法拉第常数	Faraday's constant	96485 C mol ⁻¹
理想气体常数	Gas constant	8.3145 J K ⁻¹ mol ⁻¹
电子质量	Mass of an electron	9.1094×10 ⁻³¹ kg
光速	Speed of light	2.99792458×10 ⁸ ms ⁻¹
普朗克常数	Plank Constant	6.62637×10 ⁻³⁴ Js

朗伯-比尔定律 Beer-Lambert law $A=\epsilon lc$

1 atm = 101325 Pa
1 bar = 100000 Pa
0 °C = 273.15 K